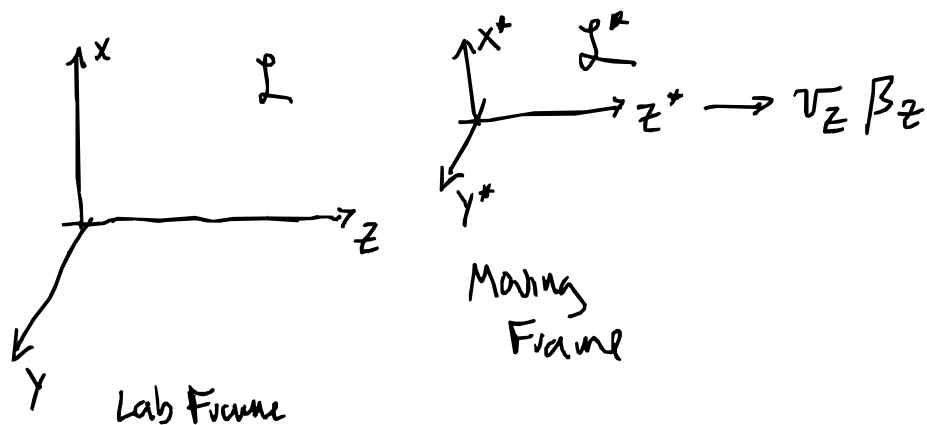


Special Relativity

Ref: Weidemann Ch. 1.4



To move between frames $\rightarrow$ Lorentz Transformations

$$x = x^*$$

$$y = y^*$$

$$z = \gamma (z^* + \beta_z ct^*)$$

$$ct = \gamma (\beta_z z^* + ct^*)$$

$$\gamma = \frac{1}{\sqrt{1 - \beta_z^2}}$$

Lorentz Factor

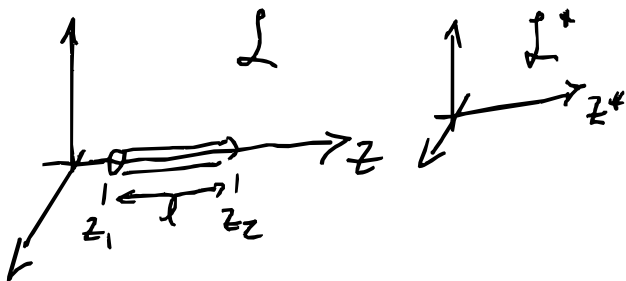
$$\beta_z = v_z/c$$

$\uparrow$ Not Beta function!

In matrix formalism:

$$\begin{bmatrix} x \\ y \\ z \\ ct \end{bmatrix} = M_L \begin{bmatrix} x^* \\ y^* \\ z^* \\ ct^* \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \gamma & \beta\gamma \\ 0 & 0 & \beta\gamma & \gamma \end{bmatrix} \begin{bmatrix} x^* \\ y^* \\ z^* \\ ct^* \end{bmatrix}$$

How is length measured in a moving frame?



$$\begin{aligned} l &= z_2 - z_1 \\ &= \gamma(z_2^* + v \cancel{t_2^*}) - \gamma(z_1^* + v \cancel{t_1^*}) \end{aligned}$$

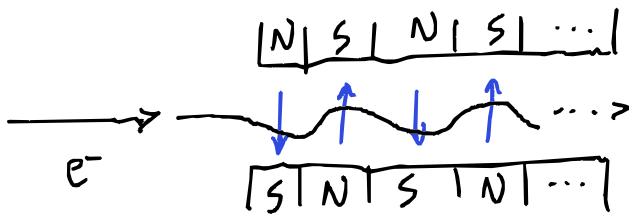
$$\text{Note: } t^* = t_2^* = t_1^*$$

$$= \gamma(z_2^* - z_1^*)$$

$$\Rightarrow \boxed{\begin{aligned} l^* &= \frac{l}{\gamma} \\ v^* &= \frac{v}{\gamma} \end{aligned}}$$

$\Rightarrow$ length contraction!

Example: Magnetic Undulator



$\lambda_p \equiv$ period of the undulator

$\therefore$ In the e^- frame of reference

$$\lambda_p^* = \frac{\lambda_p}{\gamma}$$

$\rightarrow$ This is how a relativistic e^- can radiate coherently in the X-ray regime

What about the fields?

$$\begin{bmatrix} E_x \\ E_y \\ E_z \\ cB_x \\ cB_y \\ cB_z \end{bmatrix} = \begin{bmatrix} \gamma & 0 & 0 & 0 & \gamma\beta_z & 0 \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \end{bmatrix} \begin{bmatrix} E_x^* \\ E_y^* \\ E_z^* \\ cB_x^* \\ cB_y^* \\ cB_z^* \end{bmatrix}$$

$$\rightarrow E_x = \gamma E_x^* + \gamma\beta_z cB_y$$

Static $B_y \rightarrow$ EM wave in e^- frame

Lorentz invariance + 4-vectors

Space-time
4-vector: $A^\mu = (x, y, z, ct)$

Property of 4-vectors is that the length of a 4-vector is a Lorentz invariant

$$\therefore A^\mu A_\mu = x^2 + y^2 + z^2 - c^2 t^2 = \text{const.}$$

$$\rightarrow c d\tau = \sqrt{1 - \beta^2} c dt$$

↑
Lorentz invariant
world line

$$\rightarrow d\tau = \frac{1}{\gamma} dt \Rightarrow \text{time dilation}$$

Velocity
4-vector:
$$U^\mu = \frac{d}{d\tau} A^\mu = \gamma \frac{d}{dt} A^\mu$$
$$= c \gamma (\beta_x, \beta_y, \beta_z, 1)$$

Momentum

4-vector:
$$P^\mu = m U^\mu = m c \gamma (\beta_x, \beta_y, \beta_z, 1)$$

Force
4-vector

$$\vec{F} = m\vec{a} = \frac{d\vec{p}}{dt}$$

$$\rightarrow F^\alpha = \frac{dp^\alpha}{dt}$$

For charged particles $\rightarrow$ we care most about
the Lorentz Force

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$\nwarrow$ acceleration $\nwarrow$ Bending/Focusing

In full 4-vector notation:

$$F^\alpha = \frac{dp^\alpha}{dt} = \frac{q}{c} F^{\alpha\beta} U_\beta$$

$\nearrow$ velocity $\searrow$ Fields

Finish with a derivation:

Start with the momentum-energy 4-vector:

$$cp^\alpha = (cp_x, cp_y, cp_z, iE)$$

Remember $\rightarrow$ invariant length of a 4-vector
is a const. in all frames

$$c^2 p^\alpha p_\alpha = c^2 p^2 - E^2 = m^2 c^4 \quad \text{Rest mass!}$$

$$\left. \begin{array}{l} \text{in Rest frame: } p \rightarrow 0 \\ E \rightarrow E_0 \end{array} \right\}$$

So, in any frame:

$$\boxed{E^2 = c^2 p^2 + m^2 c^4} \quad (1)$$

Similarly - multiply 2 diff. 4-vectors this time
velocity $\times$ momentum 4-vector

$$(\gamma \vec{v}, i\gamma c)(c\vec{p}, iE)$$

$$= \gamma \vec{v} c \vec{p} - \gamma c E$$

$$= -\gamma_0' mc^3 \quad \text{in Rest frame}$$

$$p \rightarrow 0$$

$$E \rightarrow E_0$$

$$\gamma \rightarrow \gamma_0 = 1$$

$$\rightarrow \gamma \vec{v} c \vec{p} - c \gamma E = -mc^3$$

$$\text{Note: } \vec{v} \parallel \vec{p}$$

$$\gamma \beta c p = \gamma E - mc^2$$

$$\boxed{cp = \frac{\gamma E - mc^2}{\gamma \beta}} \quad (2)$$

Now $\rightarrow$ put ① and ② together:

$$\begin{aligned} E^2 &= c^2 p^2 + m^2 c^4 \\ &= \left(\frac{\gamma E - mc^2}{\gamma \beta} \right)^2 + m^2 c^4 \end{aligned}$$

$$\text{Note: } \beta^2 \gamma^2 = \gamma^2 - 1$$

$$\rightarrow E^2 = \frac{\gamma^2 E^2 - 2\gamma E mc^2 + m^2 c^4}{\gamma^2 - 1} + m^2 c^4$$

$$(E^2 - m^2 c^4)(\gamma^2 - 1) = \gamma^2 E^2 - 2\gamma E mc^2 + m^2 c^4$$

$$\cancel{\gamma^2 E^2} - E^2 - \gamma^2 m^2 c^4 + \cancel{m^2 c^4} = \cancel{\gamma^2 E^2} - 2\gamma E mc^2 + \cancel{m^2 c^4}$$

$$E^2 - 2\gamma E mc^2 + \gamma^2 m^2 c^4 = 0$$

$$(E - \gamma mc^2) = 0$$

$$\Rightarrow \boxed{E = \gamma mc^2}$$

